Feature Engineering Criteria: Student Pass/Fail Clustering

Assignment overview

Your task is to clean the student data, add a few new features, and check whether K-Means groups pass and fail students better afterwards.

Dataset columns (600 students):

Column	Meaning	Range
study_hours_per_week	Self-study hours per week	0–24
attendance_pct	Class attendance	52–100 %
previous_grade	Grade in the previous course	24–100
assignments_done_pct	Share of assignments submitted	35–100 %
sleep_hours	Average sleep per night	4–10 h
absences	Number of absences	0–19
actual	True outcome (Pass / Fail)	**Evaluation only**

Deliverables:

1. Your code (script or notebook) that creates the new dataset.
2. Answer sheet to demonstrate the solution which include Accuracy before and after feature engineering, with 2–3 sentences explaining the result. (*Accuracy, Silhouette score, Confusion Matrix*) (See the answer sheet sample)

Two rules to follow

1. **Remove the `actual` column before you start.** Keep it aside and use it only at the end to check accuracy.
2. **Never build a feature from the `actual` column.** The model must not see the answer.

Steps to follow

Follow these five steps in order.

Step	What to do	How to check
1. Clean	Remove duplicate rows and fill empty cells with the column median	`df.duplicated().sum()` and `df.isna().sum()` both show 0
2. Fix wrong values	Make sure percentages are between 0 and 100 and hours are not negative	`df.describe()` shows sensible min and max values

Step	What to do	How to check
3. Add new features	Create **2 new features** (see the next section)	Each new column has a one-sentence reason
4. Remove repeated features	If two features are almost the same (correlation above 0.85), keep only one	Look at `df.corr()`
5. Scale	Put all features on the same scale before K-Means	Use `StandardScaler()`

Why scaling matters: K-Means measures distance. Without scaling, a feature with big numbers (attendance 0–100) would matter much more than one with small numbers (sleep 4–10).

Bonus (optional): apply `np.log1p()` to a skewed column such as `absences`, and say whether it helped.

Making a new feature

A good new feature passes two simple tests:

1. **It makes sense:** you can explain in one sentence why it could affect passing the exam.
2. **It doesn't use the answer:** it is made only from the feature columns, never from `actual`.

Ideas you can use (pick any 2, or invent your own):

New feature	How to make it	Why it might help
engagement_score	(attendance_pct + assignments_done_pct) / 2	Students who attend and do homework tend to pass
sleep_deviation	abs(sleep_hours – 8)	Too little or too much sleep can hurt results
risk_count	Count how many are true: study < 2 h, attendance < 70 %, previous grade < 50, absences ≥ 10	More warning signs, higher risk of failing

Checking your result

Run K-Means (k = 2, `random_state=42`) twice: once on the original features, once on your new dataset. Fill in this table:

	Before	After
Accuracy (compared with `actual`)		

	Before	After
Silhouette score (higher = clearer groups)		

Then write 2–3 sentences: did it improve, and which feature do you think helped most? A result that did not improve is fine if you explain it.

Grading (100 points)

Part	Points	Full marks if…
Cleaning (steps 1–2)	20	Duplicates, empty cells and wrong values are checked
No answer leakage	20	`actual` is removed before engineering and used only for accuracy
2 new features	30	Both features work and each has a sensible reason
Remove repeats and scale (steps 4–5)	10	Correlation is checked and **StandardScaler** is used
Before/after table and explanation	20	Table is complete and the explanation makes sense

Common mistakes

- Leaving the `actual` column in the data when running K-Means.
- Forgetting to scale the features.
- Adding a feature that is almost a copy of an existing one.
- Reporting accuracy without explaining what changed.